Design and Analysis of Algorithm

Lecture 24: Trees

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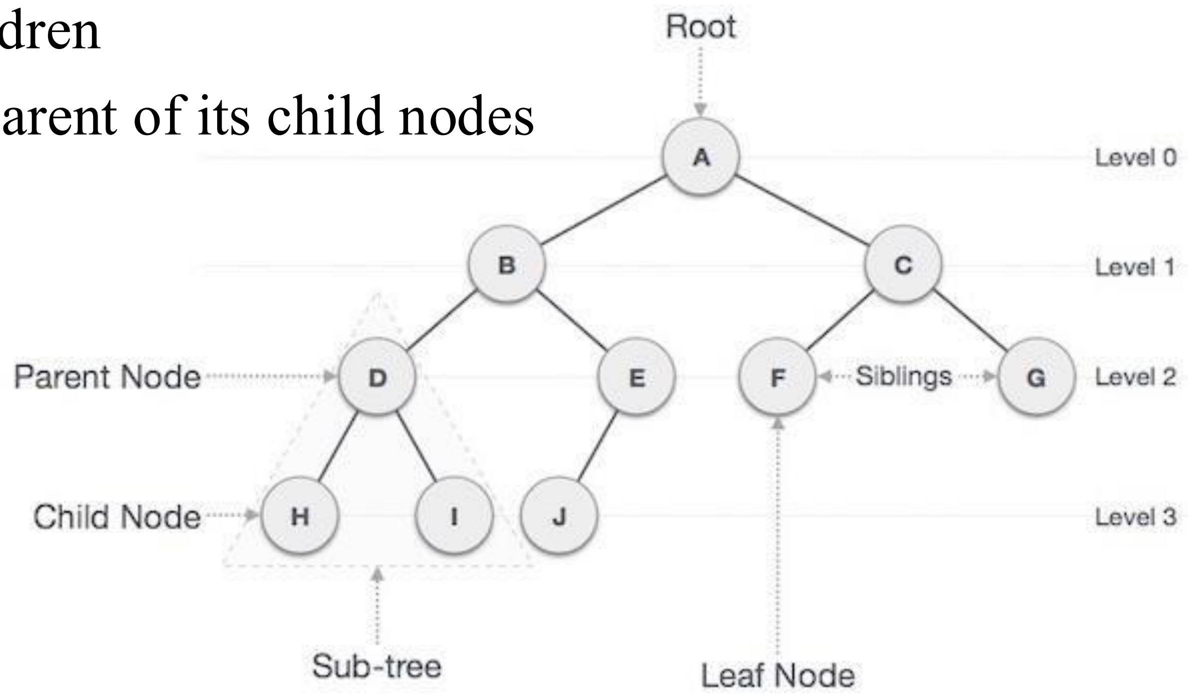
- Introduction

Tree: Introduction

A tree is an abstract data type with following characteristics

- One entry point, the root
- Each node is either a leaf or an internal node
- An internal node has 1 or more children
- The internal node is said to be the parent of its child nodes



Tree: Keywords

Siblings: two nodes that have the same parent

Edge: the link from one node to another

Path Length: the number of edges that must be traversed to get from one node to another

Depth: the path length from the root of the tree to this node

height of a node: The maximum distance of any leaf from this node

- a leaf has a height of 0
- the height of a tree is the height of the root of that tree

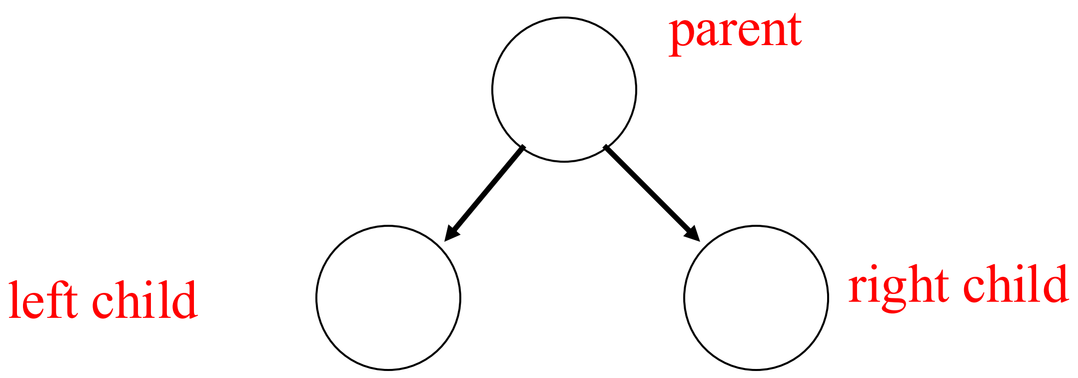
Descendants: any nodes that can be reached via 1 or more edges from this node

Ancestors: any nodes for which this node is a descendant

Binary Tree

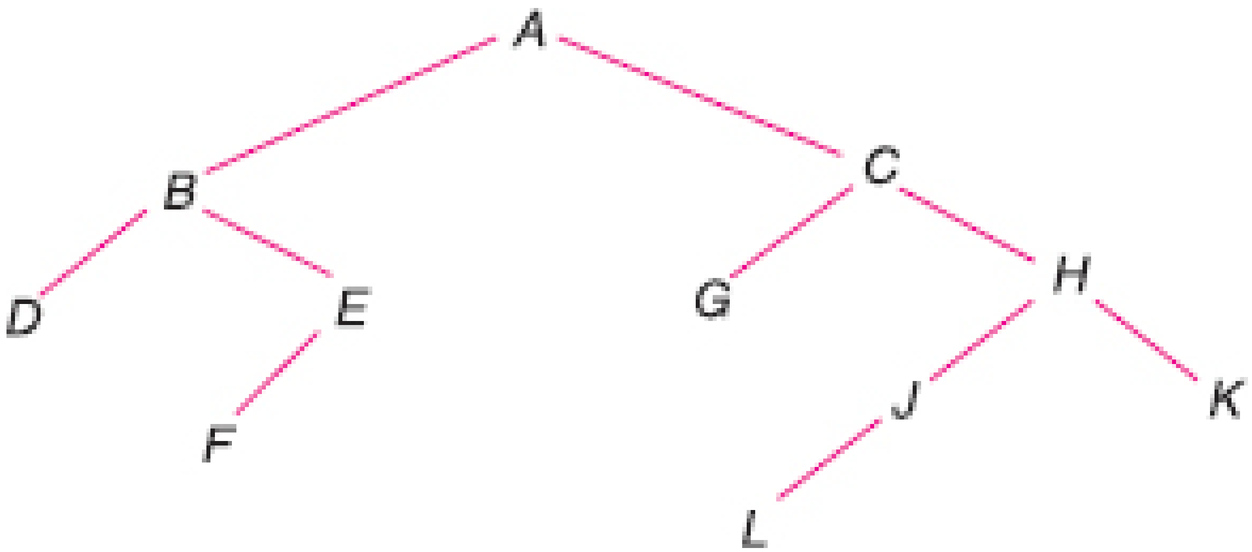
A binary tree T is defined as a finite set of elements, called nodes, such that:

- T is empty (called the null tree or empty tree), or
- T contains a distinguished node R, called the root of T, and the remaining nodes of T form an ordered pair of disjoint binary trees T_1 and T_2
- If T_1 is nonempty, then its root is called the left successor of R;
- If T_2 is nonempty, then its root is called the right successor of R.



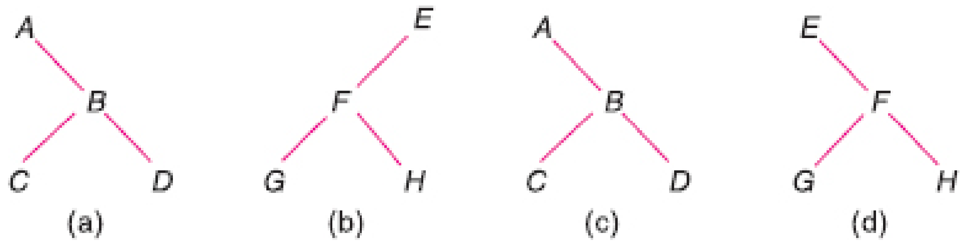
In binary tree each node has at most two children

Example: 1



- B is a left successor and C is a right successor of the node A.
- The left subtree of the root A consists of the nodes B, D, E and F, and
- The right subtree of A consists of the nodes C, G, H, J, K and L.

Similar Trees

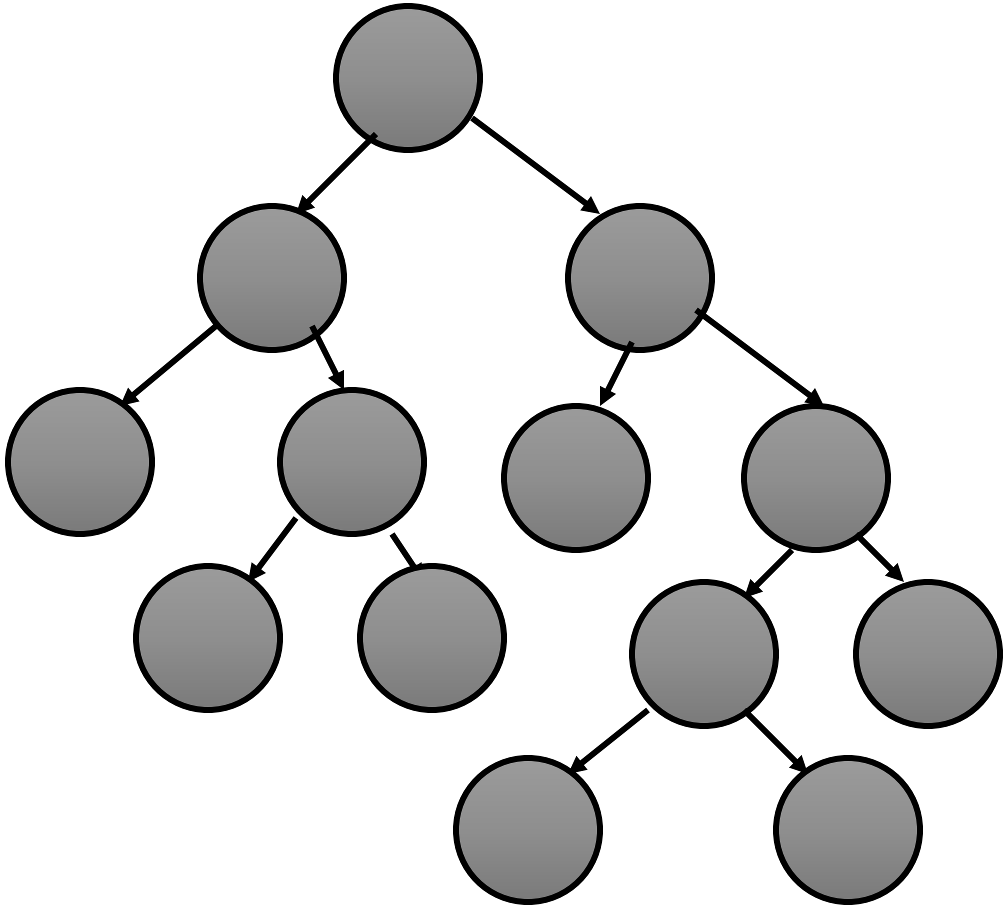


- Similar Trees: a) , c) d)
- Tree Copies: a) and c)
- Non-Similar: b)

Full Binary Tree

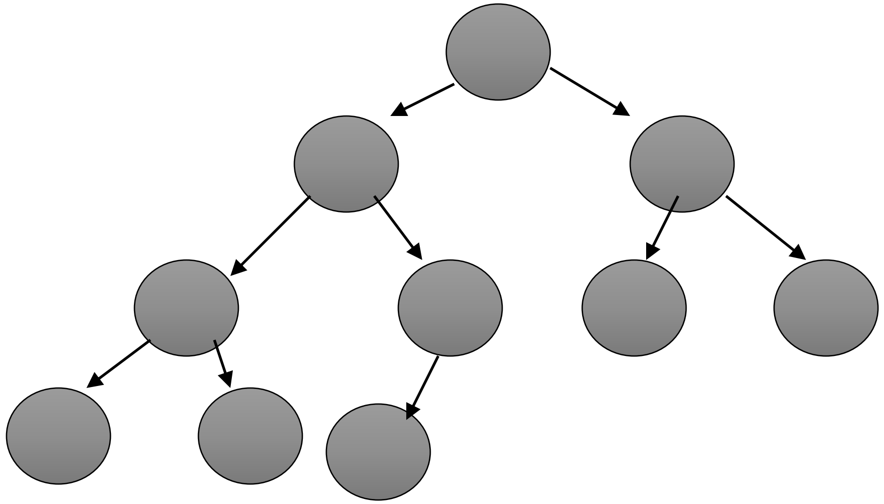
It is also called, 2-Tree or Extended binary Tree

A binary tree in which each node has 2 or 0 children



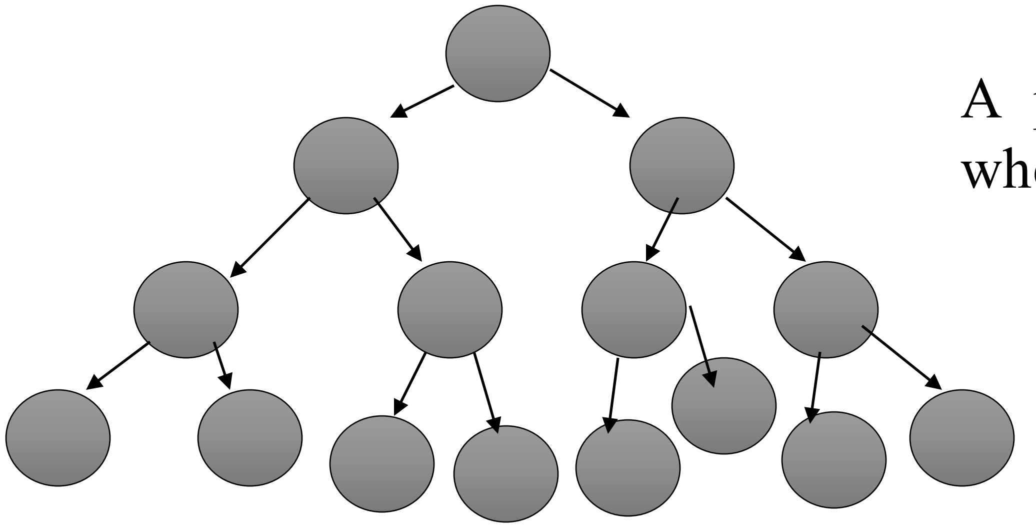
Complete Binary Tree

A binary tree in which every level, except possibly the deepest is completely filled



Perfect Binary Tree

A binary tree with all leaf nodes at the same depth. All internal nodes have exactly two children.



A perfect binary tree has $2^{h+1} - 1$ nodes where h is the height of the tree

Note: A perfect binary tree has the maximum number of nodes for a given height

Questions

1. What is the height of a complete binary tree that contains N nodes?

    1. O(1)
    2. O(logN) 2
    3. O($N^{1/2}$)
    4. O(N)
    5. O(NlogN)

2. If the complete tree T_n has $n = 1\ 000\ 000$ nodes, then its depth is

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Binary Tree Representation

There are two different ways of representing trees in memory

- The first way uses a single array, and is called the sequential representation of binary tree.
- The second way is called the link representation of binary tree and is analogous to the way linked lists are represented in memory.

Binary Tree: Sequential Representation

This representation uses only a single linear array (say TREE)

Rules to determine the locations of the parent, left child and right child of any node I in the binary tree using array.

1. Store the root node in the first location of the array TREE
2. If a node is present in the location I of the array TREE then store its left child in the location $(2 * I)$
3. If a node is present in the location I of the array TREE then store its right child in the location $(2 * I + 1)$.

Left, Right and Root of tree

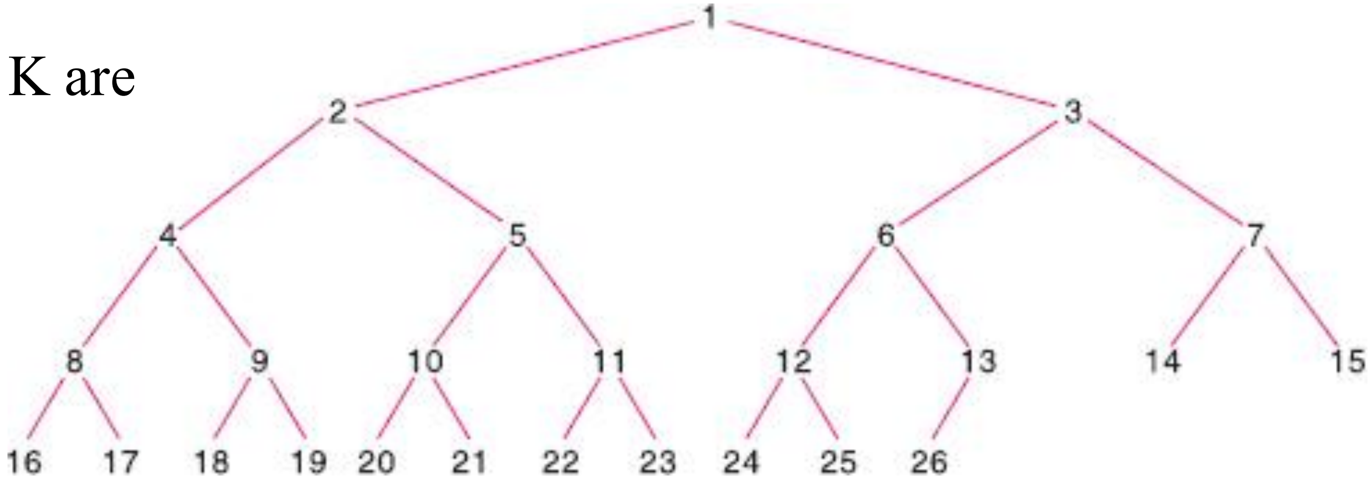
Consider that the index number of the nodes are given as marked in the image.

The left and right children of the node K are

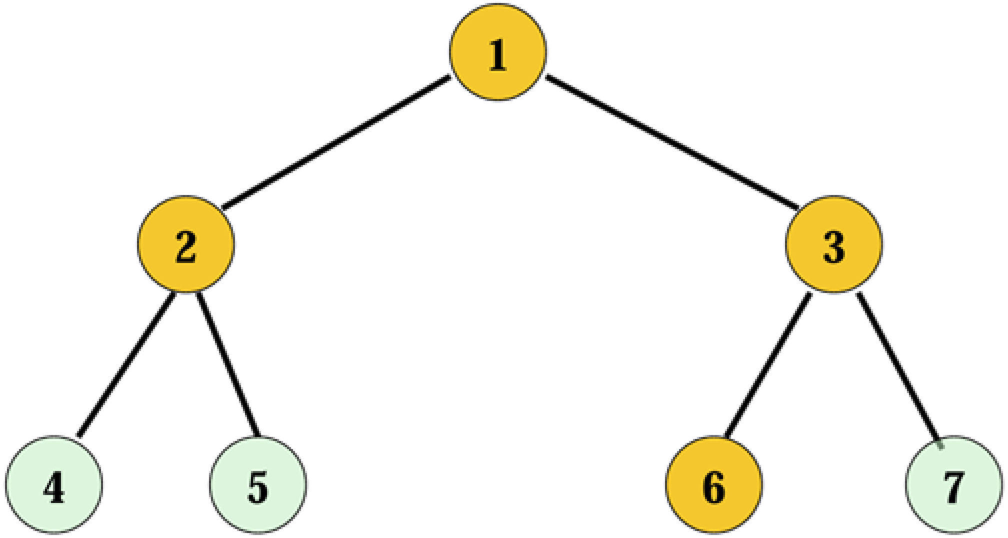
Left:$2K$ Right: $2k + 1$

The root of node K is

$$\lfloor K/2 \rfloor$$



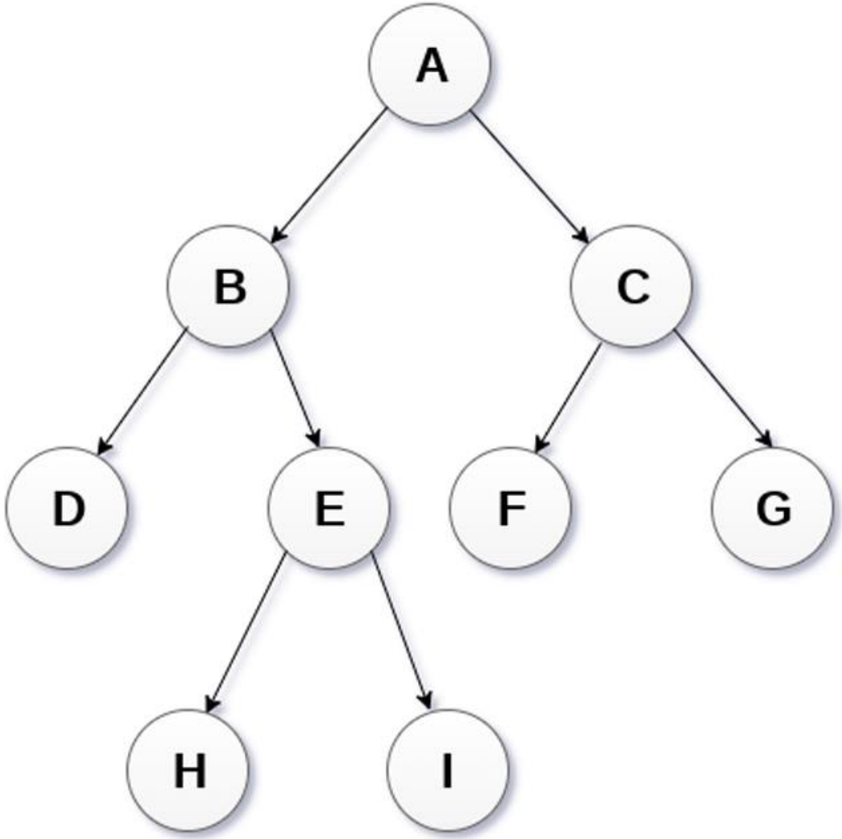
Example 2



0	1	2	3	4	5	6
1	2	3	4	5	6	7

Question

Represent the following binary tree in an array

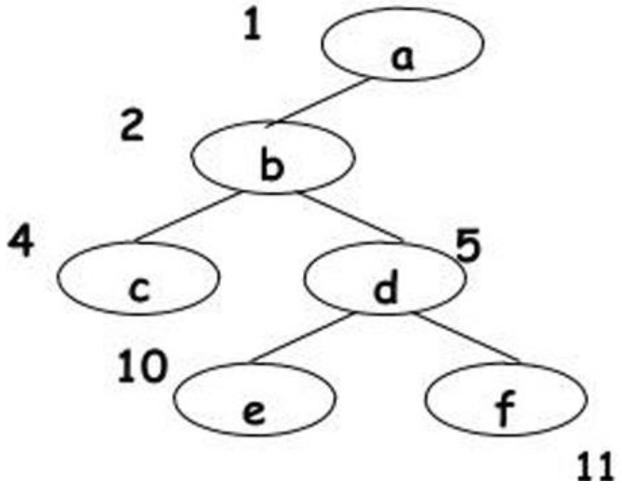


A	B	C	D	E	F	G			H	I
1	2	3	4	5	6	7	8	9	10	11

Question

What is the approximate length of array required to represent a binary tree of depth d

1. $Log_2 d + 1$
2. $Log_2(d + 1)$
3. $2^d + 1$
4. 2^{d+1}



1	2	3	4	5	6	7	8	9	10	11
a	b		c	d					e	f

Advantage and disadvantage

Advantages:

- This representation is very easy to understand.
- This is the best representation for complete and full binary tree representation.
- Programming is very easy.
- It is very easy to move from a child to its parents and vice versa.

Disadvantages:

- Lot of memory area wasted.
- Insertion and deletion of nodes needs lot of data movement.
- Execution time high.
- This is not suited for trees other than full and complete tree.

Linked representation

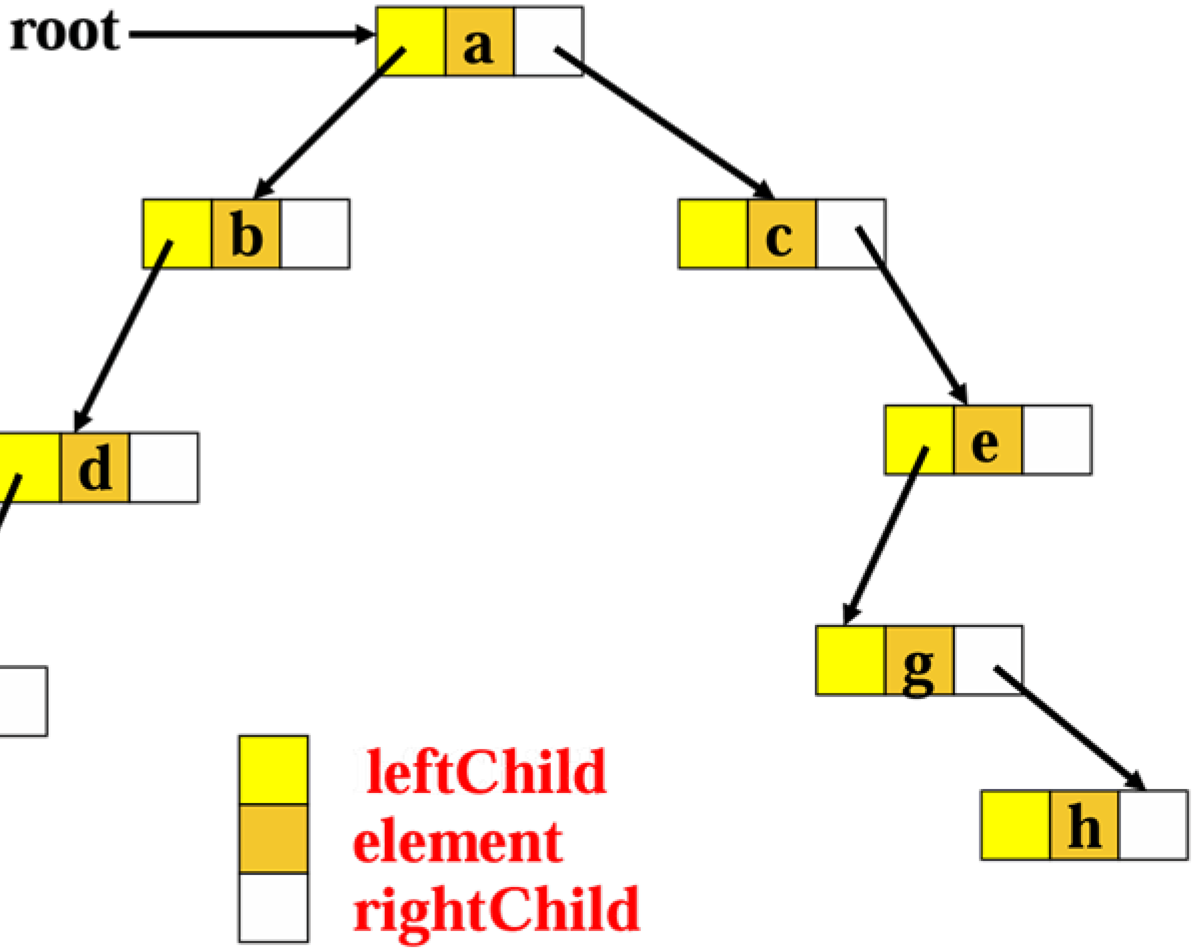
In this representation, the binary tree is stored in the memory, in the form of a linked list where the number of nodes are stored at non contiguous memory locations and linked together by inheriting parent child relationship like a tree.

Every node contains three parts : pointer to the left node, data element and pointer to the right node.

Each binary tree has a root pointer which points to the root node of the binary tree.

In an empty binary tree, the root pointer will point to null.

Linked representation: Example



Advantage and disadvantage

Advantages:

- No data movement required for Insertion and Deletion
- No movement of nodes except the rearrangement of pointers

Disadvantages:

- Given a node structure, it is difficult to determine its parent node
- Wastage of memory space in storing NULL pointer for the nodes which have no subtrees